



PRECISION FORGED THERMAL VISION LASER FOCUS

We are a high-volume manufacturing factory that integrates innovative R&D with state-of-the-art production, offering a one-stop solution for all your infrared thermal imaging and laser rangefinder needs.

As a leading manufacturer in the industry, we focus on designing and producing various advanced thermal imaging and laser rangefinder products. From tactical applications to commercial solutions, our technology is designed for performance, durability, and precision.

Thermal Camera

Uncooled long-wave infrared



Detector Type	Vanadium Oxide Uncooled Infrared Focal Plane Detector		
Resolution	256×192	384×288	640×512
Effective Frame Rate	25/50Hz	30/60Hz	30/60Hz
Pixel Pitch	12μm		
Spectral Band	8~14μm		
Noise Equivalent	≤40mK@25°C, F#1.0, 25Hz		
Thermal Time Constant	<12ms		
Analog Video	PAL/NTSC		
Digital Video	DVP/BT656/USB2.0/MIPI		
Communication Interface	UART/I2C/USB2.0		
Weight (Without Lens)	<7g		<8.6g
Dimensions (Without Lens)	21×21×10.3mm		
Operating Temperature	-40°C~+80°C		
Lens	4/9.1/13.5/18/35/55mm		



Detector Type	Vanadium Oxide Uncooled Infrared Focal Plane Detector	
Resolution	640×512	
Effective Frame Rate	30/60Hz	
Pixel Pitch	10μm	
Spectral Band	8~14μm	
Noise Equivalent	≤40mK@25°C, F#1.0, 25Hz	
Thermal Time Constant	<12ms	
Analog Video	PAL/NTSC	
Digital Video	DVP/BT656/ USB2.0/MIPI	
Communication Interface	UART/I2C/USB2.0	
Weight (Without Lens)	<4.3g	
Dimensions (Without Lens)	14.5mm×14.5mm×12mm	
Operating Temperature	-40°C~+80°C	
Lens	11mm	



Detector Type	Vanadium oxide uncooled infrared Focal plane detector
Resolution	640×512
Effective Frame Rate	60Hz
Pixel Pitch	8μm
Spectral Band	8~14μm
Noise Equivalent	≤40mK@25°C, F#1.0, 25Hz
Thermal Time Constant	15ms
Analog Video	PAL/NTSC
Digital Video	USB2.0
Communication Interface	UART/I2C/USB2.0
Weight (Without Lens)	9g
Dimensions (Without Lens)	21×21×10.3mm
Operating Temperature	-30°C~+70°C
Lens	6mm

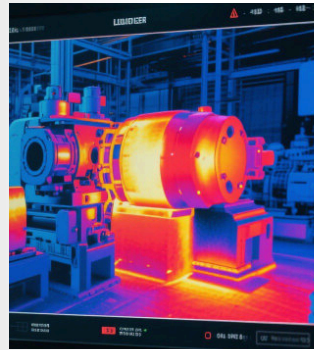


Detector Type	Vanadium Oxide Uncooled Infrared Focal Plane Detector
Resolution	1280×1024
Effective Frame Rate	Imaging 50Hz(Thermometry 30Hz)
Pixel Pitch	12μm
Spectral Band	8~14μm
Noise Equivalent	≤40mK@25°C, F#1.0, 25Hz
Thermal Time Constant	<12ms
Analog Video	/
Digital Video	DVP/BT1120/MIPI/USB3.0
Communication Interface	UART/I2C
Weight (Without Lens)	<26g
Dimensions (Without Lens)	≤29×29mm
Operating Temperature	Imaging: -40°C~+80°C Thermometry: -20°C~+60°C
Lens	19/35/55mm



Detector Type	Vanadium Oxide Uncooled Focal Plane Detector
Effective Pixels	640x512
Pixel Pitch	12μm
NETD	s35mK@25C, F/1.0
Frame Rate (fps)	USB:50Hz, CVBS:25Hz
Spectral Range	8~14μm
Field of view (FOV)	47.7°(H)×38.2°(V) 32.3(H)°×26.1°(V) 61.5°(D) 40.6°(D)
Dimensions (Without Lens)	17.3x17.3x19.5mm
Weight (Without Lens)	7.5g
Video Output	UVC/CVBS
Hardware Interfaces	USB2.0/CVBS
Operating Temperature	-40°C to +70°C (-40°F to +158°F)
Noise Reduction	2D NR/3D NR
Lens	9.1/13mm

Applications Scenarios



Industrial predictive maintenance



Monitoring



Fire and rescue operations



Scientific research and ecology



Consumer electronics integration

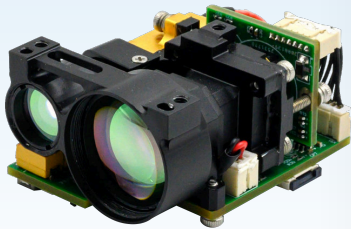


Technology advantages in key markets

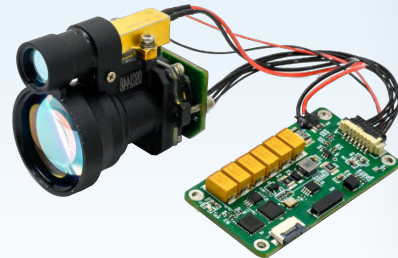


Contact
Information

Laser Rangefinder



LRF-4Km	
Wavelength	1.535μm±0.01μm
Divergence	0.4mrad±0.1mrad
Max Range	Buildings: ≥4km
Min Range	≤30m
Frequency	1~10Hz
Accuracy	≤±1m
Detection Probability	≥98%
Range-Finding Mode And Time	≥60min
Supply Voltage	3.3V~16V
Power Consumption	Operating Power: ≤3W Standby Power: ≤1.5W
Interface	RS422/TTL
Dimension	≤48.5×36×26mm
Weight	≤60g
Operation Temperature	-40°C~+70°C
Storage Temperature	-50°C~+75°C



LRF-6Km	
Wavelength	1.535μm±0.01μm
Divergence	0.4mrad±0.1mrad
Max Range	Buildings: ≥6km
Min Range	≤30m
Frequency	1~10Hz
Accuracy	≤±2m
Detection Probability	≥98%
Range-Finding Mode And Time	≥30min
Supply Voltage	3.3V~16v
Power Consumption	Operating Power: ≤3.5W Standby Power: ≤1.5W
Interface	RS422/TTL
Dimension	≤55×34×45mm
Weight	≤75g
Operation Temperature	-40°C~+70°C
Storage Temperature	-50°C~+75°C



LRF-8Km	
Wavelength	1535nm±5nm
Divergence	≤0.5mrad
Max Range	Buildings: ≥8km
Min Range	≤30m
Frequency	1~10Hz
Accuracy	≤±1m
Detection Probability	≥98%
Range-Finding Mode And Time	≥30min
Supply Voltage	9V~15V
Power Consumption	Operating Power: ≤3.5W Standby Power: ≤1.5W
Interface	RS422
Dimension	≤51×35×45mm
Weight	≤75g
Operation Temperature	-40°C~+60°C
Storage Temperature	-50°C~+70°C



LRF-10Km	
Wavelength	1535nm±5nm
Divergence	≤0.3mrad
Max Range	Buildings: ≥10 km
Min Range	≤30m
Frequency	1~10Hz
Accuracy	≤±1m
Detection Probability	≥98%
Range-Finding Mode And Time	≥30min
Supply Voltage	9V~15V
Power Consumption	Operating Power: ≤3.5W Standby Power: ≤1.5W
Interface	RS422
Dimension	≤68×42.5×50mm
Weight	≤100g
Operation Temperature	-40°C~+60°C
Storage Temperature	-50°C~+70°C



LRF-20Km	
Wavelength	1535nm±5nm
Divergence	≤0.5mrad
Max Range	Buildings: ≥20km
Min Range	≤100m
Frequency	1~10Hz
Accuracy	≤±3m
Detection Probability	≥98%
Range-Finding Mode And Time	≥30min
Supply Voltage	22V~34V
Power Consumption	≤10W
Interface	RS422
Dimension	132x94x102.7mm
Weight	840g±20g
Operation Temperature	-40°C~+60°C
Storage Temperature	-50°C~+70°C

Applications
Scenarios



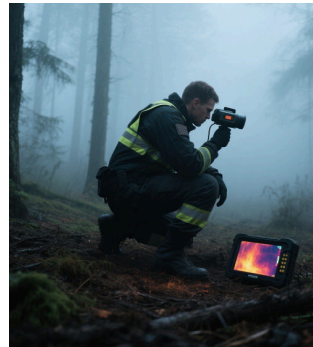
Intelligent Vehicle Mobility



Autonomous navigation
sensor



Polar/plateau expeditions



Forestry resource survey



Power line inspection



Bridge monitoring